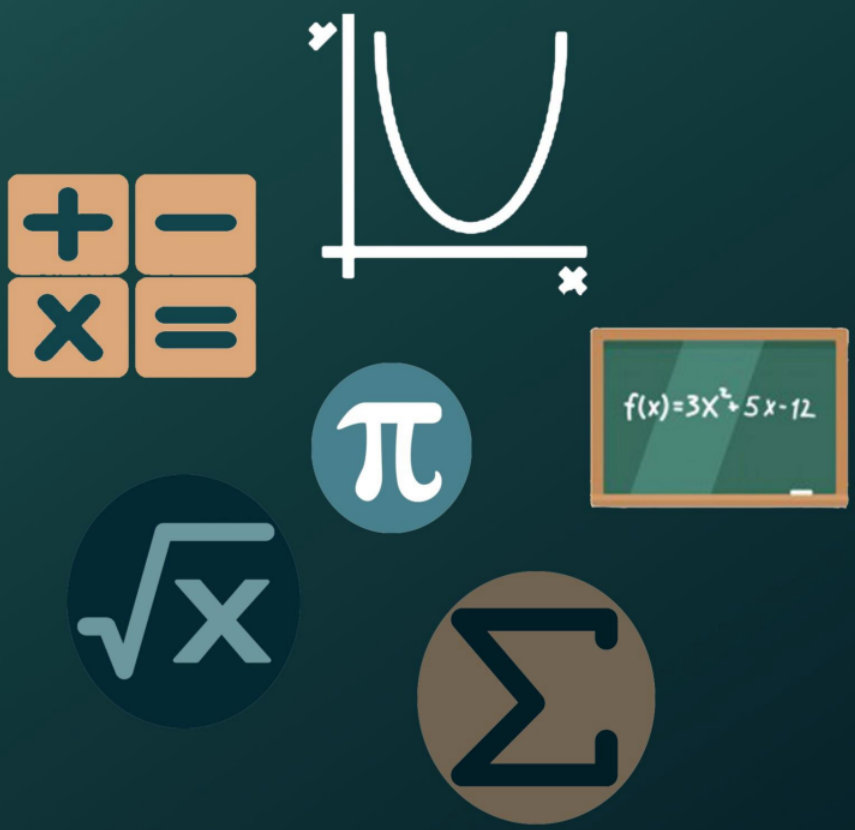
MATHEMATICS

Trigonometric Equation



SHEET

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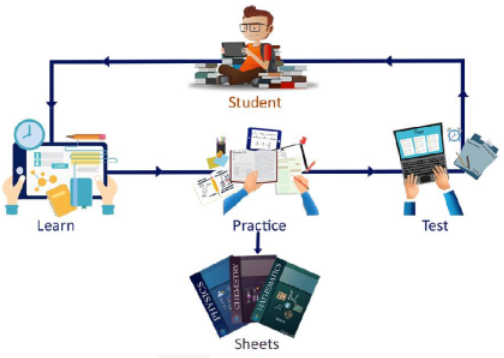
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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr. No.	Topic	Ex - 1	Ex - 1A	Ex - 2	Ex - 2A
1	Introduction to Trigonometric Equations				
2	General Solution of Trigonometric Equations - Part 1	1-6	1		1
3	General Solution of Trigonometric Equations - Part 2	7-9	7-8		
4	Strategies for Solving Trigonometric Equations - Type 1, 2	10-15	2, 14-15	1	2, 19-20
5	Strategies for Solving Trigonometric Equations - Type 3	16-18	3-4	2	21
6	Strategies for Solving Trigonometric Equations - Type 4, 5	19-20	17, 19	3-4	3, 22
7	Strategies for Solving Trigonometric Equations - Type 6				4
8	Strategies for Solving Trigonometric Equations - Type 7, 8, 9	21-25	16, 18	5-10	5-10, 23-26
9	Trigonometric Inequalities & System of Inequality	26-29	5-6	11-16	27
10	After Complete Chapter	30-31	9-13	17-21	11-18

Exercise - 1 Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

GENERAL SOLUTION OF TRIGONOMETRIC EQUATIONS - PART 1

Q1. The general solution of $\sin 3\theta = 0$ is :-

(A) $n\pi; n \in I$ (B) $\frac{n\pi}{3}; n \in I$

(C) $(2n+1)\frac{1}{2}; n \in I$ (D) $2n\pi; n \in I$

Q2. $\cos\theta = \frac{1}{2}$ then θ is-

(A) $2n\pi \pm \frac{\pi}{6}$ (B) $n\pi \pm \frac{\pi}{2}$

(C) $2n\pi \pm \frac{\pi}{3}$ (D) $n\pi$

Q3. General solution of equation $\frac{\tan 4x - \tan 2x}{1+\tan 4x \tan 2x} = 1$ is :-

(A) $n\pi + \frac{\pi}{4}$ (B) $\frac{n\pi}{2} + \frac{\pi}{8}$

(C) $n\pi$ (D) None of these

Q4. General solution of $\tan 5\theta = \cot 2\theta$ is (where $n \in I$) :-

(A) $\theta = \frac{n\pi}{7} + \frac{\pi}{14}$ (B) $\theta = \frac{n\pi}{7} + \frac{\pi}{5}$

(C) $\theta = \frac{n\pi}{7} + \frac{\pi}{2}$ (D) $\theta = \frac{n\pi}{7} + \frac{\pi}{3}$

Q5. If $\tan\theta + \tan 4\theta + \tan 7\theta = \tan\theta \tan 4\theta \tan 7\theta$, then $\theta =$

(A) $\frac{n\pi}{4}$ (B) $\frac{n\pi}{7}$

(C) $\frac{n\pi}{12}$ (D) $n\pi$

Q6. The number of solutions of the equation $\frac{\sec x}{1-\cos x} = \frac{1}{1-\cos x}$ in $[0, 2\pi]$ is equal to -

(A) 3 (B) 2

(C) 1 (D) 0

GENERAL SOLUTION OF TRIGONOMETRIC EQUATIONS - PART 2

Q7. If $\sin^2\theta = \frac{1}{4}$, then the general solution of θ.

(A) $2n\pi \pm \frac{\pi}{6}$ (B) $n\pi \pm \frac{\pi}{4}$

(C) $n\pi \pm \frac{\pi}{6}$ (D) $2n\pi \pm \frac{\pi}{3}$

Q8. The general solution of equation $4\cos^2 x + 6\sin^2 x = 5$ is-

(A) $x = n\pi \pm \frac{\pi}{2}$ (B) $x = n\pi \pm \frac{\pi}{4}$

(C) $x = n\pi \pm \frac{3\pi}{2}$ (D) None of these

Q9. If $2\tan^2\theta = \sec^2\theta$, then the general solution of θ-

(A) $2n\pi \pm \frac{\pi}{6}$ (B) $n\pi \pm \frac{\pi}{4}$

(C) $n\pi \pm \frac{\pi}{6}$ (D) $2n\pi \pm \frac{\pi}{3}$

STRATEGIES FOR SOLVING TRIGONOMETRIC EQUATIONS - TYPE 1, 2

Q10. The equation $3\sin^2x + 10\cos x - 6 = 0$ is satisfied if :-

(A) $x = n\pi \pm \cos^{-1}\left(\frac{1}{3}\right)$

(B) $x = 2n\pi \pm \cos^{-1}\left(\frac{1}{3}\right)$

(C) $x = n\pi \pm \cos^{-1}\left(\frac{1}{6}\right)$

(D) None

Q11. The complete solution of the equation $7\cos^2 x + \sin x \cos x - 3 = 0$ is given by -

(A) $n\pi + \frac{\pi}{2}; (n \in I)$

(B) $n\pi - \frac{\pi}{4}; (n \in I)$

(C) $n\pi + \tan^{-1}\frac{4}{3}; (n \in I)$

(D) $n\pi + \frac{3\pi}{4}, k\pi + \tan^{-1}\frac{4}{3}; (n, k \in I)$

Q12. The total number of solution of equation $\sin^4x + \cos^4x = \sin x \cos x$ in $[0, 2\pi]$ is equal to -

(A) 2

(B) 4

(C) 6

(D) None of these

Q13. If $\sin 2\theta = \cos 3\theta$, and θ is an acute angle then $\sin\theta$ is equal to :-

(A) $\frac{\sqrt{5}-1}{4}$ (B) $-\frac{\sqrt{5}-1}{4}$

(C) 0 (D) None

Q14. If $\cos 2x - 3\cos x + 1 = \frac{\operatorname{cosec} x}{\cot x - \cot 2x}$, then which of the following is true ?

(A) $x = (2n+1)\frac{\pi}{2}, n \in I$

(B) $x = 2n\pi, n \in I$

(C) $x = 2n\pi \pm \cos^{-1}\left(\frac{2}{5}\right), n \in I$

(D) no solution

Q15. The general solution of $\tan^2\theta + (1-\sqrt{3})\tan\theta - \sqrt{3} = 0$ is-

(A) $n\pi + \frac{\pi}{4}; n \in I$

(B) $n\pi - \frac{\pi}{3}; n \in I$

(C) $n\pi - \frac{\pi}{4}; m\pi + \frac{\pi}{3}; n, m \in I$

(D) None of these

STRATEGIES FOR SOLVING TRIGONOMETRIC EQUATIONS - TYPE 3

Q16. General solution of equation $\sqrt{3}\cos\theta + \sin\theta = \sqrt{2}$ is :-

(A) $2n\pi + \frac{5\pi}{12}$ (B) $2n\pi + \frac{\pi}{12}$

(C) $2n\pi$ (D) None of these

Q17. The number of solutions of $\cos\theta + \sqrt{3}\sin\theta = 5$, $0 \le \theta \le 5\pi$ is :-

(A) 4 (B) 0

(C) 5 (D) None

Q18. Number of solution(s) of the equation $\sin 2\theta + \cos 2\theta = -\frac{1}{2}, \theta \in \left[0, \frac{\pi}{2}\right]$, is -

(A) 0 (B) 1

(C) 2 (D) 3

STRATEGIES FOR SOLVING TRIGONOMETRIC EQUATIONS - TYPE 4, 5

Q19. If $\sin6\theta + \sin4\theta + \sin2\theta = 0$, Then θ equals to :

(A) $\frac{n\pi}{4}$ or $n\pi \pm \frac{\pi}{6}$

(B) $\frac{n\pi}{4}$ or $n\pi \pm \frac{\pi}{3}$

(C) $\frac{n\pi}{4}$ or $2n\pi \pm \frac{\pi}{6}$

(D) $\frac{n\pi}{4}$ or $2n\pi \pm \frac{\pi}{3}$

Q20. If $\cos p\theta = \cos q\,\theta$, $p \neq q$ then :-

(A) $\theta = 2n\pi$ (B) $\theta = \frac{2n\pi}{p \pm q}$

(C) $\theta = \frac{n\pi}{p+q}$ (D) None

STRATEGIES FOR SOLVING TRIGONOMETRIC EQUATIONS - TYPE 7, 8, 9

Q21. If $x, y \in [0, 2\pi]$ and $\sin x + \sin y = 2$, then value of $x + y$ is :-

(A) π (B) $\pi/2$

(C) 3π (D) None of these

Q22. The number of real solutions of the equation $\cos^5 x + \sin^3 x = 1$ in the interval $[0, 2\pi]$ is:-

(A) 2 (B) 1

(C) 3 (D) Infinite

Q23. If $x + y = \frac{2\pi}{3}$ and $\frac{\sin x}{\sin y} = 2$, Then :-

(A) $x - y = 2n\pi + \frac{\pi}{3}$ (B) $x - y = 2n\pi$

(C) $x + y = 2n\pi$ (D) $x + y = 2n\pi + \frac{\pi}{3}$

Q24. Find the general solution for x when

$\sin x\left(\cos\frac{x}{4} - 2\sin x\right) + \left(1 + \sin\frac{x}{4} - 2\cos x\right)\cos x = 0$

(A) $2p\pi$, $p \in I$ (B) $(8p-6)\pi$, $p \in I$

(C) $(3p-2)\pi$, $p \in I$ (D) $(2p+1)\pi$, $p \in I$

Q25. Find the general solution for x when $\sin 3x + \cos 2x = -2$

(A) $(4p-3)\frac{\pi}{3}$, $p \in I$

(B) $(4p-2)\frac{\pi}{3}$, $p \in I$

(C) $(4p+1)\frac{\pi}{7}$, $p \in I$

(D) None of these

TRIGONOMETRIC INEQUALITIES & SYSTEM OF INEQUALITY

Q26. If $\cos\theta - \sin\theta = \sqrt{2}\sin\theta$ then $\cos\theta + \sin\theta$ can be equal to :-

(A) $\sqrt{2}\cos\theta$ (B) $-\sqrt{2}\sin\theta$

(C) $\sqrt{2}\sin\theta$ (D) $\cos\theta$

Q27. Solution set of inequality $\cos\theta - \sqrt{3}\sin\theta \geq 1$, if $\theta \in (-\pi, \pi)$

(A) $(-\pi, 0)$ (B) $(0, \pi)$

(C) $[-\pi/3, \pi/3]$ (D) $\left[\frac{-2\pi}{3}, 0\right]$

Q28. The solution of inequality $\sin x \leq \cos x$ in $[0, 2\pi]$ is :-

(A) $\left(0, \frac{\pi}{4}\right) \cup \left(\frac{5\pi}{4}, 2\pi\right)$

(B) $\left[\frac{\pi}{4}, \frac{5\pi}{4}\right]$

(C) $\left[0, \frac{\pi}{4}\right] \cup \left[\frac{5\pi}{4}, 2\pi\right]$

(D) $\left[0, \frac{\pi}{2}\right] \cup \left[\frac{3\pi}{2}, 2\pi\right]$

Q29. Set of values of x lying in $[0, 2\pi]$ satisfying the inequality $|\sin x| > 2\sin^2 x$ contains

(A) $\left(0,\frac{\pi}{6}\right)\cup\left(\pi,\frac{7\pi}{6}\right)$

(B) $\left(0,\frac{7\pi}{6}\right)$

(C) $\pi/6$

(D) $\left(\frac{\pi}{6},\pi\right)$

Q30. If $\tan a\theta - \tan b\theta = 0$, then the values of θ form a series in –

(A) A.P. (B) G.P.

(C) H.P. (D) None of these

Q31. Find the number of solutions of the equation: $30|\sin x| = x$ in $0 \le x \le 2\pi$

(A) 4 (B) 2

(C) 8 (D) 6

Exercise - 1A *Concept Builder*

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. $\cos\left(\frac{3\theta}{2}\right)=0$; General value(s) of θ are $(n \in I)$:-

(A) $(2n+1)\frac{\pi}{3}$ (B) $(4n\pm1)\frac{\pi}{3}$

(C) $(6n\pm1)\frac{\pi}{3}$ (D) $(8n\pm1)\frac{\pi}{3}$

Q2. If $4\cos^2\theta+\sqrt{3}=2(\sqrt{3}+1)\cos\theta$, then θ can be equal to-$(n\in I)$-

(A) $2n\pi\pm\frac{\pi}{3}, n \in I$

(B) $2n\pi\pm\frac{\pi}{4}, n \in I$

(C) $2n\pi\pm\frac{\pi}{6}, n \in I$

(D) $(6n\pm1)\frac{\pi}{3}$

Q3. If $\sqrt{3}\sin\theta - \cos\theta = \sqrt{2}$, then θ is equal to $(n\in I)$:-

(A) $2n\pi - \frac{\pi}{3} \pm \frac{3\pi}{4}$

(B) $n\pi + (-1)^n \frac{\pi}{4}+\frac{\pi}{6}$

(C) $2n\pi + \frac{\pi}{3} + \frac{3\pi}{4}$

(D) $n\pi + (-1)^n \frac{\pi}{4}-\frac{\pi}{6}$

Q4. Solution of equation $1 + \cot\theta = \operatorname{cosec}\theta$ is $(n\in I)$-

(A) $2n\pi \pm \frac{\pi}{4}+\frac{\pi}{4}$ (B) $n\pi+(-1)^n\frac{\pi}{4}-\frac{\pi}{4}$

(C) $2n\pi + \frac{\pi}{2}$ (D) None of these

Q5. Which of the following are correct solutions for $\tan x \ge \sqrt{3}$:-

(A) $\left[\frac{\pi}{3},\frac{\pi}{2}\right)$ (B) $\left(\frac{\pi}{2},\frac{2\pi}{3}\right]$

(C) $\left[\frac{4\pi}{3},\frac{3\pi}{2}\right)$ (D) $\left[\frac{7\pi}{3},\frac{5\pi}{2}\right)$

Q6. Which of the following are possible values of x for the equation $\sin 2x - \sin x > 0$:-

(A) $\left(\frac{\pi}{3}, \pi\right)$ (B) $\left(0, \frac{\pi}{3}\right)$

(C) $\left(-\pi, \frac{-7\pi}{6}\right)$ (D) $\left(-\pi, \frac{-\pi}{2}\right)$

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Passage # 1 (Ques. 7 to 8)

Consider the equation $5\cos^2\theta + 7\sin^2\theta - 6 = 0$

Q7. If $\theta \in [0, 2\pi]$ Number of solutions are :-

(A) 1 (B) 2

(C) 3 (D) 4

Q8. If $\theta \in [0, 2\pi]$ sum of all possible values of $\sin\theta$ are :-

(A) 0 (B) 1

(C) –1 (D) 2

Passage # 2 (Q. 9 to 11)

If $P_n = \sin^n\theta + \cos^n\theta$ where $n \in N$ then answer the following :

Q9. If $P_1 = m$ then value of $4(1 - P_6)$ is -

(A) $3(m-1)^2$ (B) $3(m^2-1)^2$

(C) $3(m^2+1)^2$ (D) none

Q10. Value of $2P_6 - 3P_4 + 10$ is-

(A) 0 (B) 6

(C) 9 (D) none

Q11. If $P_{n-2} - P_n = \sin^2\theta \cos^2\theta P_\lambda$ then value of λ is -

(A) n – 1 (B) n – 2

(C) n – 3 (D) n – 4

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q12.

Column-I (Trigonometric Equations)	**Column-II** (General Solution)
(A) $2\sin x - \sqrt{3} = 0$	(P) $n\pi + (-1)^n \frac{\pi}{3}$
(B) $\sin 2\theta + \cos 2\theta + 4\sin\theta = 1 + 4\cos\theta$	(Q) $2n\pi + \frac{2\pi}{3}$
(C) $\sqrt{3}\sin x - \cos x = 2$	(R) $n\pi + (-1)^n \frac{\pi}{10}$
(D) $4\cos^2 x \sin x - 2\sin^2 x = 3\sin x$	(S) $n\pi + \frac{\pi}{4}$
	(T) $n\pi + (-1)^n \frac{\pi}{12}$

MATCHING LIST TYPE

*Each question has TWO (02) **matching lists: LIST I and LIST II**. FOUR options are given representing matching of elements from LIST I and LIST II. ONLY ONE of these four options corresponds to a correct matching. For each question, choose the option corresponding to the correct matching.*

Q13. **List– I**

(P) $\cot\left(\frac{\pi}{4}-\frac{x}{3}\right)=\frac{\sqrt{3}}{3}$

(Q) $\sin x\sin 7x = \sin 3x\sin 5x$

(R) $\sin^3 x - \cos^3 x = 0$

(S) $2\tan^2 x+3=\frac{3}{\cos x}$

List – II

(1) $\frac{\pi(4n+1)}{4}; n\in I$

(2) $\frac{n\pi}{4}; n\in I$

(3) $\frac{\pi(12n-1)}{4}; n\in I$

(4) $2n\pi ; n\in I$

Codes :

	P	Q	R	S
(A)	3	4	1	2
(B)	1	4	1	2
(C)	3	2	1	4
(D)	2	3	1	4

SUBJECTIVE TYPE

Q14. Solve for x ; $2\sin^2 2x + 6\sin^2 x = 5$

Q15. Solve for x ; $\sin 3x + \sin^3 x = \frac{3\sqrt{3}}{4}\sin 2x$

Q16. Find the number of solutions of $\tan x + \sec x = 2\cos x$ in $[0, 2\pi]$.

Q17. Solve : $\cos\theta + \cos 3\theta + \cos 5\theta + \cos 7\theta = 0$

Q18. Solve for x and y : $2^{\frac{1}{\cos^2 x}}\sqrt{y^2-y+1/2}\le 1$

Q19. Find the number of solutions for $\sin 5\theta \cos 3\theta = \sin 9\theta . \cos 7\theta$ in $\left[0,\frac{\pi}{2}\right]$.

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. Let

$$f(x)=\frac{\operatorname{cosec}^4x-2\operatorname{cosec}^2x+1}{\operatorname{cosec}x(\operatorname{cosec}x-\sin x)+\frac{\sin x-\cos x}{\sin x}+\cot x}.$$

The sum of all the solution of f (x) = 0 in [0, 100π] is -

(A) 2550π. (B) 2500π.

(C) 5000π. (D) 5050π.

Q2. Find the number of distinct solutions of secx + tanx = $\sqrt{3}$, where $0 \le x \le 3\pi$.

(A) 2 (B) 3

(C) 4 (D) 5

Q3. If $\frac{\sin 6x}{\sin x}=8\cos x\cos 2x\cos 4x$,

then general solution for x is -

(A) $x = n\pi,\ n \in I$

(B) $x=(2n+1)\frac{\pi}{2},\ n\in I$

(C) $x=(2n+1)\frac{\pi}{14},\ n\in I$

(D) None

Q4. Number of solutions of the trigonometric equation in [0, π], $\sin 3\theta = 4\sin\theta.\sin 2\theta.\sin 4\theta$

(A) 4 (B) 6

(C) 8 (D) 10

Q5. The general solution of equation

$2^{\cos 2x}=3.2^{\cos^2 x}-4$ is :-

(A) nπ ± π/6 (B) 2nπ ± π/3

(C) nπ ± π/2 (D) nπ

Q6. Which of the following is not solution of equaiton $2^{\cos 2x}+1=3.2^{-\sin^2 x}$:-

(A) nπ, n ∈ z

(B) $\left(n+\frac{1}{2}\right)\pi, n\in z$

(C) $\left(n-\frac{1}{2}\right)\pi, n\in z$

(D) $\left(n\pm\frac{1}{4}\right)\pi, n\in z$

Q7. No. of solution of equation 5 sin x = |x| is:-

(A) 3 (B) 4

(C) 2 (D) 5

Q8. Number of solution of 8cosx = x will be :-

(A) 3 (B) 4

(C) 5 (D) 6

Q9. The values of x satisfying the system of equations $2^{\sin x+\cos y}=1$, $16^{\sin^2 x+\cos^2 y}=4$ are given by-

(A) $x = n\pi+(1)^n\frac{\pi}{6}$ and $y = 2n\pi\pm\frac{\pi}{3}$

(B) $x = n\pi+(-1)^{n+1}\frac{\pi}{6}$ and $y = 2n\pi\pm\frac{2\pi}{3}$

(C) $x = n\pi+(-1)^n\frac{\pi}{6}$ and $y = 2n\pi\pm\frac{2\pi}{3}$

(D) $x = n\pi+(-1)^{n+1}\frac{\pi}{3}$ and $y = 2n\pi\pm\frac{\pi}{6}$

Q10. The expression $(1 + \tan x + \tan^2 x)(1 - \cot x + \cot^2 x)$ has the positive value for x given by :-

(A) $0 \le x \le \pi/2$

(B) $0 \le x \le \pi$

(C) $x \in R$

(D) $x \ge 0$

Q11. Solution set of inequality $2\cos^2\theta + \sin\theta \geq 2$ if $\theta \in \left[\frac{\pi}{2}, \frac{3\pi}{2}\right]$:-

(A) $\left[\frac{\pi}{2}, \frac{5\pi}{6}\right] \cup \left[\pi, \frac{3\pi}{2}\right]$ (B) $\left[\frac{\pi}{2}, \frac{5\pi}{6}\right]$

(C) $\left[\pi, \frac{3\pi}{2}\right]$ (D) $\left[\frac{5\pi}{6}, \pi\right]$

Q12. The number of values of $\theta \in [0, 4\pi]$ satisfying the $\left|\sqrt{3}\cos x - \sin x\right| \geq 2$ is :-

(A) 0 (B) 2 (C) 4 (D) 8

Q13. The number of solution of the equation $|\cos x - \sin x| = 2\cos x$ in $[0, 2\pi]$ is :-

(A) 1 (B) 2 (C) 3 (D) 4

Q14. Complete set of values of x in $(0, \pi)$ satisfying $1 + \log_2 \sin x + \log_2 \sin 3x \geq 0$, is -

(A) $\left(\frac{\pi}{6}, \frac{\pi}{4}\right] \cup \left[\frac{3\pi}{4}, \frac{5\pi}{6}\right]$

(B) $x \in \left(\frac{2\pi}{3}, \frac{3\pi}{4}\right)$

(C) $\left(0, \frac{\pi}{2}\right) \cup \left(\frac{2\pi}{3}, \pi\right)$

(D) $\left(\frac{\pi}{2}, \frac{2\pi}{3}\right)$

Q15. The number of integral values of a for which the equation $\cos 2x + a \sin x = 2a - 7$ possesses a solution is -

(A) 2 (B) 3 (C) 4 (D) 5

Q16. Find complete set of all negative values of 'a' which makes the quadratic inequality $\sin^2 x + a\cos x + a^2 \geq 1 + \cos x$ true for every $x \in R$.

(A) $a \in (-\infty, -2]$ (B) $a \in (-\infty, -3)$

(C) $a \in (-\infty, -4)$ (D) None of these

Q17. Which of the following can be a root of the equation; $\sin(x - 2) = \sin(3x - 4)$ in $(-\pi, \pi)$.

(A) $-\frac{7\pi}{2} + \frac{3}{2}$ (B) $-\frac{5\pi}{4} + \frac{3}{2}$

(C) $\frac{-3\pi}{4} + \frac{3}{2}$ (D) None of these

Q18. If

$$\tan^2[\pi(x+y)] + \cot^2[\pi(x+y)] = 1 + \sqrt{\frac{2x}{1+x^2}}$$

where $x, y \in R$, then least positive value of y is-

(A) $\frac{5}{4}$ (B) $\frac{1}{4}$ (C) $\frac{3}{4}$ (D) 2

Q19. Solve for x, $\sum_{r=1}^{n} \sin(rx) \cdot \sin(r^2x) = 1$

(A) $\frac{(2m+1)\pi}{n(n-1)}$ (B) $\frac{(4m-1)\pi}{(n+1)}$

(C) $\frac{(2m+1)\pi}{n(n+1)}$ (D) None of these

Q20. x_1 and x_2 are two positive values of x for which $2\cos x$, $|\cos x|$ and $3\sin^2 x - 2$ are in G.P. The possible value of $|x_1 - x_2| + \cos^{-1}\frac{2}{3} = \frac{p\pi}{q}$ then $(q - p)$ is equal to (p,q are co-prime numbers-

(A) 0 (B) 1

(C) 2 (D) 3

Q21. If $(\cos x)^{\sin^2 x - 3\sin x + 2} = 1$, then all solutions of x are given by :

(A) $2n\pi + \frac{\pi}{2}$ (B) $(2n+1)\pi - \frac{\pi}{2}$

(C) $2n\pi$ (D) $n\pi$

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. If $\sin\left(\frac{6}{5}x\right) = 0$ and $\cos\left(\frac{x}{5}\right) = 0$, then

(A) $x = (n-5)\pi$ (B) $x = 6(n-1)\pi$

(C) $x = 5\left(n-\frac{1}{2}\right)\pi$ (D) $x = 5\left(n+\frac{1}{2}\right)\pi$

Q2. Root of the equation $2\sin^2\theta + \sin^2 2\theta = 2$ is:

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$ (C) $\frac{5\pi}{4}$ (D) $\frac{\pi}{2}$

Q3. If $\cos^2 x + \cos^2 2x + \cos^2 3x = 1$ then -

(A) $x = (2n+1)\frac{\pi}{4}, n \in I$

(B) $x = (2n+1)\frac{\pi}{2}, n \in I$

(C) $x = n\pi \pm \frac{\pi}{6}, n \in I$

(D) None of these

Q4. Which of the following set of values of x satisfies the equation

$2^{(2\sin^2 x - 3\sin x + 1)} + 2^{(2 - 2\sin^2 x + 3\sin x)} = 9$.

(A) $x = n\pi \pm \frac{\pi}{6}, n \in I$

(B) $x = n\pi \pm \frac{\pi}{3}, n \in I$

(C) $x = n\pi, n \in I$

(D) $x = 2n\pi + \frac{\pi}{2}, n \in I$

Q5. For a given value of k the number of different solutions of the equation $3\cos\theta + 4\sin\theta = k$ in the range $0^\circ \le \theta \le 360^\circ$ are :-

(A) zero if $|k| > 5$

(B) two if $|k| < 5$

(C) one if $|k| = 5$

(D) no solution for any k

Q6. $\cos(x-y) - 2\sin x + 2\sin y = 3$ if -

(A) $\sin x = \sin y$

(B) $x + y = 2n\pi$, $x - y = (2k+1)\pi$

(C) $x = 2k\pi - \frac{\pi}{2}, y = 2n\pi + \frac{\pi}{2}$

(D) $\cos(x-y) = -1$ $(n, k \in I)$

Q7. If $x = X\cos\theta - Y\sin\theta, y = X\sin\theta + Y\cos\theta$ and $x^2 + 4xy + y^2 = AX^2 + BY^2, 0 \le \theta \le \frac{\pi}{2}$ then -

(a) $\theta = \pi/6$ (b) $\theta = \pi/4$

(c) $A = 3$ (d) $B = 1$

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Passage # 1 (Ques. 8 to 10)

Consider the system of equations
$\sin x \cos 2y = (a^2 - 1)^2 + 1$ and $\cos x \sin 2y = a + 1$.

Q8. Number of values of a for which the system has a solution is -

(A) 1 (B) 2

(C) 3 (D) Infinite

Q9. Number of values of $x \in [0, 2\pi]$ when the system has solution for permissible values of a -

(A) 1 (B) 2

(C) 3 (D) 4

Q10. Number of values of $y \in [0, 2\pi]$ when the system has solution for permissible values of a -

(A) 2 (B) 3

(C) 4 (D) 5

Passage # 2 (Ques. 11 to 13)

Consider the cubic equation

$$x^3 - (1+\cos\theta+\sin\theta)x^2 + (\cos\theta\sin\theta+\cos\theta+\sin\theta)x - \sin\theta.\cos\theta = 0$$

Whose roots are x_1, x_2 and x_3.

Q11. The value of $x_1^2 + x_2^2 + x_3^2$ equals :

(A) 1

(B) 2

(C) $2\cos\theta$

(D) $\sin\theta(\sin\theta+\cos\theta)$

Q12. Number of values of θ in $[0,2\pi]$ for which at least two roots are equal

(A) 3 (B) 4

(C) 5 (D) 6

Q13. Greatest possible difference between two of the roots if $\theta \in [0,2\pi]$ is

(A) 2 (B) 1

(C) $\sqrt{2}$ (D) $2\sqrt{2}$

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q14.

Column - I	**Column - II**
(A) Number of roots of the equation $\cos^7 x + \sin^4 x = 1$ in the interval $[0,2\pi]$ is	(P) 1
(B) The value of 'a' for which the equation $a^2 - 2a + \sec^2\pi(a+x) = 0$ has solution is	(Q) 0
(C) Number of solutions of the equation $\lvert\cos x\rvert = 2[x]$ is (where [x] is greatest integer function)	(R) 4

Q15. The solution of

Column - I	**Column - II**
(A) $\sin x \cos^3 x > \cos x \sin^3 x$, $0 \le x \le \frac{\pi}{2}$ is	(P) $\left[-\pi, \frac{-3\pi}{4}\right] \cup \left[\frac{-\pi}{4}, \frac{\pi}{4}\right] \cup \left[\frac{3\pi}{4}, \pi\right]$
(B) $4\sin^2 x - 8\sin x + 3 \le 0$ is	(Q) $\left[\frac{3\pi}{2}, 2\pi\right] \cup \{0\}$
(C) $\lvert\tan x\rvert \le 1$ and $x \in [-\pi, \pi]$ is	(R) $(0, \pi/4)$
(D) $\cos x - \sin x \ge 1$ and $0 \le x \le 2\pi$ is	(S) $\left[\frac{\pi}{6}, \frac{5\pi}{6}\right]$

TRIPLE MATCH TYPE

*This section contains ONE table having 3 columns and 4 rows. Based on the table each question has FOUR options [A], [B], [C] AND [D]. **ONLY ONE** of these four is correct.*

Column-I contains the trigonometric equations

Column-II contains the number of solutions of x when $\frac{\pi}{4} \le x \le 2\pi$

Column-III contains the general solution of x

Column-I	**Column-II**	**Column-III**
(I) $1 + \sin x + \sin^2 x + \sin^3 x + \ldots\ldots \infty = 4 + 2\sqrt{3}$	(i) 4	(P) $n\pi \pm \frac{\pi}{3}; n \in I$
(II) $\sin 3\alpha = 4\sin\alpha(\sin(x + \alpha).\sin(x - \alpha))$	(ii) 2	(Q) $n\pi + (-1)^n \frac{\pi}{6}; n \in I$
(III) $\cos 6x + \tan^2 x + \cos 6x \tan^2 x = 1$	(iii) 1	(R) $n\pi + (-1)^n \frac{\pi}{3}; n \in I$
(IV) $2\log_{25}(5^{2\sin x} - 4) = 2\sin x + \sin\frac{3\pi}{2}$	(iv) 6	(S) $\frac{n\pi}{4}; x \ne (2n+1)\frac{\pi}{2}; n \in I$

Q16. Which of the following is the only correct combination ?
(A) (II), (i), (R) (B) (III), (iii), (P) (C) (II), (ii), (Q) (D) (II), (i), (P)

Q17. Which of the following is the only correct combination ?
(A) (IV), (i), (P) (B) (I), (ii), (Q) (C) (III), (iv), (S) (D) (I), (iv), (S)

Q18. Which of the following is the any incorrect combination ?
(A) (IV), (iii), (Q) (B) (III), (iv), (S) (C) (I), (ii), (R) (D) (I), (ii), (Q)

SUBJECTIVE TYPE/ NUMERICAL TYPE

Q19. Solve : $2\sin^2 x - 5\sin x \cos x - 8\cos^2 x = -2$.

Q20. Solve: $\tan x + \tan 2x + \tan 3x = 0$.

Q21. Solve :
$3\sqrt{3}\sin^3 x + \cos^3 x + 3\sqrt{3}\sin x \cos x = 1$

Q22. Solve : $\cos\theta \cos 2\theta \cos 3\theta = \frac{1}{4}$;
where $0 \le \theta \le \pi$.

Q23. Find the number of solution of the equation $[0, 2\pi]$, $\tan(5\pi\cos\alpha) = \cot(5\pi\sin\alpha)$.

Q24. Prove that the equation
$k\cos x - 3\sin x = k + 1$
possess a solution iff $k \in (-\infty, 4]$.

Q25. The number of solution(s) of
$2\cos^2\left(\frac{x}{2}\right)\sin^2 x = x^2 + \frac{1}{x^2}$, $0 \le x \le \pi/2$,
is/are -

Q26. Solve the equation for x & y
$\sin^4 x = 1 + \cos^6 y$

Q27. Find the values of x in the interval $[0,2\pi]$ which satisfy the inequality :
$3|2\sin x - 1| \ge 3 + 4\cos^2 x$.

Answer Key

Ex-1									
1	2	3	4	5	6	7	8	9	10
A	C	D	A	C	D	C	B	B	B
11	12	13	14	15	16	17	18	19	20
D	A	A	D	C	A	B	B	B	B
21	22	23	24	25	26	27	28	29	30
A	C	A	B	D	A	D	C	A	A
31									
A									

Ex - 1A

1. (A,B) **2.** (A,C) **3.** (A,B) **4.** A, B, D **5.** A, C, D

6. (B,D) **7.** D **8.** A **9.** B **10.** C

11. D **12.** A→P; B→S; C→Q; D→R **13.** C **14.** $x = (2n+1)\left(\frac{\pi}{4}\right)$

15. $x = n\pi$, $x = (2m+1)\frac{\pi}{2}$ & $x = 2p\pi \pm \frac{\pi}{6}$ **16.** 2

17. $\theta = (2n_1 + 1)\,\pi/2$, $n_1 \in I$, $\theta = (2n_2 + 1)\frac{\pi}{4}$, $n_2 \in I$ or $\theta = (2n_3 + 1)\frac{\pi}{8}$, $n_3 \in I$

18. $x = n\pi$, $n \in I$ and $y = 1/2$ **19.** 9

Ex-2									
1	2	3	4	5	6	7	8	9	10
C	A	C	C	D	D	B	C	C	C
11	12	13	14	15	16	17	18	19	20
D	C	B	A	D	A	C	B	C	B
21									
D									

Ex - 2A

1. C, D **2.** B, C, D **3.** A, B, C **4.** A, D **5.** A, B, C

6. B, C, D **7.** B, C **8.** A **9.** B **10.** D

11. B **12.** C **13.** A

14. A→R; B→P; C→ Q **15.** A → R, B → S, C → P, D → Q

16. D **17.** C **18.** D

19. $x = n\pi + \alpha$ or $n\pi + \beta$ where $\alpha = \tan^{-1} 2$ & $\beta = \tan^{-1}\left(\frac{-3}{4}\right)$

20. $x = \frac{n\pi}{3}$ or $m\pi \pm \tan^{-1}\left(\frac{1}{\sqrt{2}}\right)$, $n, m \in I$

21. $x = 2n\pi$ **or** $x = 2n\pi + \frac{2\pi}{3}$ **22.** $\theta = \frac{\pi}{8}, \frac{\pi}{3}, \frac{3\pi}{8}, \frac{5\pi}{8}, \frac{2\pi}{3}, \frac{7\pi}{8}$ **23.** 28

24. Solution not required **25.** No Solution exists

26. $x = (2n+1)\frac{\pi}{2}$, $y = (2m+1)\frac{\pi}{2}$ **27.** $x \in \left[\frac{7\pi}{6}, \frac{11\pi}{6}\right] \cup \left\{\frac{\pi}{2}\right\}$

-: Important Notes & Formulas :-